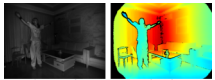
five sensing streams (one training clip: jumping jacks)

IR + depth video — 640 x 480, 16 frames



thermal video — 320 x 240, 16 frames



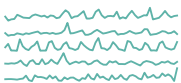
heat of the body and of objects

skeleton — 17 joints, 3-D, per frame



tracked joints in a gravity-aligned world frame

five inertial units — wrists, ankles, chest



radar — session key, point cloud



YOLO11n crops

specialists (accuracy alone, OOF)

person crop, 384 px
MViTv2-S, Kinetics init
21.6 MB — **0.733**

wrist crop, 224 px
MViTv2-S, Kinetics init
21.6 MB — **0.720**

thermal frame, 224 px
MViTv2-S, Kinetics init
21.6 MB — **0.634**

skeleton + IMU ensemble
20 members, int8
18.9 MB — **0.618**

IMU statistics forest
ExtraTrees, 545 features
1.6 MB — **0.377**

fused 0.781

prior-corrected log-linear fusion

$$\sum_i w_i \log p_i$$

coefficients from nested cross-validation

decoded 0.830

ordered transition decoder

activity-script prior per recording group; label-free

recording group and clip order

40
everyday activities

96.37 MB
one file, every weight

unseen
subjects at test time

0.877
private accuracy
14th of 307 teams

0 / 405
predictions changed over 5 GPU generations

accuracies: pooled out-of-fold, 2,700 training clips of held-out subjects. private: Kaggle leaderboard, 204 clips of subjects never seen in training.